FHIA-11 — Is Medicine a Science or an Art?

Will AI Improve Healthcare Decisions—or Just Make Them Faster?

Introduction

Medicine has long been described as both a science and an art. More than 150 years ago, French physician Armand Trousseau observed:

"The worst man of science is he who is never an artist, and the worst artist is he who is never a man of science."

A physician relying solely on scientific knowledge may overlook the patient's individual circumstances and concerns. Conversely, compassion and intuition without a sound scientific foundation may lead to ineffective or even harmful treatments. Effective medical practice requires both scientific evidence and the empathy and clinical judgment to apply it to individual patients.

Advances in medical science have dramatically improved physicians' ability to diagnose and treat disease. Yet applying that knowledge depends on the information available at the time of the decision.

A physician may have access to sophisticated diagnostic tools, clinical guidelines, and medical research, but still lack a complete patient history. Diagnoses, medications, procedures, and treatment results may be scattered across healthcare organizations, with conflicting information or missing outcomes.

A trustworthy longitudinal patient history could strengthen the application of medical science by reconciling this information and making relevant history available when needed. When analyzed across populations with appropriate privacy protections, these histories could also reveal patterns in treatment effectiveness, patient outcomes, and cumulative healthcare costs.

What treatments have been most effective for patients with similar conditions, and at what cost?

AI could bring this evidence to physicians while also improving healthcare's administrative processes. But improving the efficiency of existing transactions is different from improving the decisions underlying them.

Consider prior authorization.

What if AI makes prior authorization dramatically more efficient—and healthcare costs don't decline?

AI could accelerate authorizations without determining whether the underlying treatment requirements and policies produce the best outcomes.

The larger opportunity is to use longitudinal patient histories, treatment costs, and outcomes to improve clinical decisions and the policies that influence them.

The objective is not to replace the art of medicine with technology, but to strengthen the application of medical science while preserving the physician's judgment and relationship with the patient.

1. The Difference Between Medical Knowledge and Patient Knowledge

Physicians draw on medical training, published research, clinical guidelines, and experience treating previous patients when recommending treatments. But their personal experience represents only a fraction of the patients who have received similar treatments. What if physicians could also draw on the treatment histories and outcomes of thousands of patients with comparable clinical characteristics?

Even understanding an individual patient's prior treatments can be challenging. Did another provider prescribe the same treatment? Was it effective? How long did the benefits last? Were there complications?

Too often, answering these questions depends on patients recalling treatments and outcomes from years earlier. Much of the treatment information exists in healthcare systems, but it is fragmented across organizations. More importantly, treatment results may never be documented.

If a patient improves and does not return, the treating physician or specialist may never learn that the treatment succeeded. If treatment fails and the patient seeks care elsewhere, the original physician may never learn that it was unsuccessful.

A trustworthy longitudinal patient history could bring together what is known, identify missing outcomes, and provide opportunities to collect them during subsequent encounters.

Across populations, these histories could support analysis of treatment effectiveness, helping answer a question that individual records cannot:

What happened to patients like me?

This opportunity extends the physician's knowledge beyond personal experience by making the collective experience of healthcare available as additional evidence for clinical decisions.

2. When Standard Treatment Requirements May Increase Healthcare Costs

Consider a patient whose physician recommends a lumbar spine MRI to investigate persistent lower-back pain.

For many patients, conservative treatment, including physical therapy, is clinically appropriate before advanced imaging. Insurers may require documentation of such treatment before authorizing an MRI. But what happens when the patient's medical history or clinical findings justify earlier imaging?

The financial implications can be significant. An analysis of hospital pricing data found that commercial insurers' median negotiated facility payment for a lumbar spine MRI was approximately $528, with the middle half of hospitals ranging from $285 to $1,453. A course of twelve physical therapy visits could also cost hundreds or more than a thousand dollars, depending on services and negotiated payment rates.

If physical therapy fails and an MRI is subsequently performed, both costs are incurred, along with the time spent waiting for a diagnosis. Conversely, successful physical therapy may eliminate the need for imaging or more expensive interventions.

The question is not whether physical therapy should precede MRI, but whether the standard requirement fits the individual patient.

A trustworthy longitudinal patient history could reveal previous unsuccessful treatments, earlier diagnostic findings, or medical conditions that justify an exception. Across populations, treatment histories, costs, and outcomes could also help evaluate whether authorization requirements produce better clinical results at lower total costs.

3. Prior Authorization: Making the Process Faster or Making Better Decisions?

Prior authorization illustrates the difference between improving transaction efficiency and improving healthcare effectiveness.

AI can reduce administrative costs for both providers and payers. Providers can automate the collection of supporting documentation and the preparation of authorization requests.

Payers can automate evaluating requests against coverage criteria, identify missing information, and speed approvals or denials.

These are valuable improvements. But faster processing does not necessarily improve the policies governing treatment decisions.

As the preceding MRI example illustrates, AI could efficiently enforce a requirement that may not fit an individual patient.

The process would be more efficient. The underlying decision might remain unchanged.

Three Complementary Applications of AI

Three distinct opportunities exist for applying AI in healthcare.

Information Reconciliation AI identifies and helps resolve conflicting, incomplete, and ambiguous information across healthcare organizations, creating trustworthy longitudinal patient histories with traceable sources. It also identifies missing treatment outcomes that may need to be collected.

Transaction AI improves the efficiency of existing healthcare activities, reducing documentation effort, accelerating authorizations, identifying coding errors, and simplifying claims processing.

Outcome-Informed AI uses trustworthy longitudinal patient histories, treatment costs, and outcomes to provide additional evidence for clinical decisions and healthcare policies.

These applications are complementary. Information Reconciliation AI establishes the reliable information foundation the others need. Transaction AI reduces administrative expense, while Outcome-Informed AI helps healthcare evaluate treatment effectiveness and improve the decisions underlying those transactions.

The larger opportunity is not simply to process healthcare decisions faster, but to improve the information and evidence behind those decisions.

4. Why Existing Healthcare Information Systems Are Not Enough

The information needed to improve healthcare decisions is distributed across providers, laboratories, imaging organizations, and payers. Existing systems document diagnoses, treatments, procedures, and payments, but the information remains fragmented, and treatment outcomes may never be recorded.

Interoperability initiatives help organizations exchange records, but exchanging information is not the same as maintaining a trustworthy longitudinal patient history.

As described in earlier FHIA papers, a national longitudinal information architecture could reconcile conflicting information, preserve source provenance, maintain treatment histories, and provide controlled access with comprehensive auditing.

Equally important, the architecture must identify missing treatment outcomes. A procedure may be documented, but did it help? How long did the improvement last? Did the patient require additional treatment?

Without these answers, healthcare can measure activity and expenditures but cannot consistently evaluate treatment effectiveness.

By assembling existing information and identifying opportunities to collect missing outcomes, FHIA could support both better-informed decisions for individual patients and privacy-protected analysis of treatment effectiveness across populations.

Resolve information once. Correct it or collect missing outcomes when necessary. Use it many times.

5. The Role of Government and the Healthcare Industry

A national longitudinal information architecture does not require a national healthcare delivery system. Providers would continue delivering care, payers would continue administering coverage and paying claims, and existing healthcare information systems would retain their operational responsibilities.

The federal government's role would be to facilitate a shared information infrastructure, much as the interstate highway system enabled more efficient transportation without replacing private transportation companies.

Rather than requiring every provider, payer, and AI developer to assemble and reconcile fragmented patient information independently, a common architecture could provide trustworthy longitudinal histories with appropriate privacy protections, access controls, and auditing.

Healthcare organizations could then develop competing clinical, administrative, and analytical AI applications using a consistent information foundation.

Conclusion

Medicine will always require both science and art. Patients differ in their medical conditions, circumstances, and preferences. Even with complete information, physicians must exercise judgment in choosing the most appropriate treatment.

But better information can reduce uncertainty. A trustworthy longitudinal patient history could help physicians apply existing medical knowledge more effectively, while treatment outcomes from comparable patients could provide additional evidence for evaluating alternatives.

The same information could help payers develop authorization policies based on demonstrated treatment effectiveness rather than relying solely on predetermined treatment requirements.

AI is already making healthcare transactions faster and less labor-intensive. These improvements are valuable, but they do not necessarily address the larger economic opportunity.

Reducing the cost of processing an authorization request is different from determining whether the treatments and policies being authorized produce the best outcomes at the lowest appropriate cost.

The objective is not to replace clinical judgment, but to strengthen it with more complete and reliable evidence.

AI's greatest contribution may not be making healthcare do the same things faster.

It may be helping healthcare learn which things it should do differently.